

BAPURAM VENKATESWAR REDDY

AI Engineer — Generative AI, RAG & Agentic Systems

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EDUCATION

National Institute of Technology Warangal (NIT Warangal)

Warangal, India

Integrated M.Sc. (B.S. + M.S.) in Mathematics

2021 – 2026

Relevant Coursework: Machine Learning, Artificial Intelligence, Data Science, Probability & Statistics, Linear Algebra, Optimization Theory, Graph Theory, Algorithms

EXPERIENCE

Devseccops AI (Sec and DevOps AI Pvt. Ltd.)

Jul 2026 – Present

Full-Stack AI Engineer Intern

- Built enterprise Generative AI and RAG applications using Python, FastAPI, LangChain, LangGraph, FAISS, Chroma, and AWS Bedrock.
- Reduced LLM inference costs by 35% through model optimization, token minimization, and prompt caching strategies.
- Designed agentic AI workflows with state, context, tools, and structured execution; reduced unnecessary LLM calls by 25%.
- Built multimodal pipelines for text, documents, speech, images, and video using Amazon Nova, Gemini, Whisper Large v3, Ray 2, and FFmpeg.
- Achieved 2x higher media-processing throughput through parallelized worker queues and async execution pipelines.
- Reduced manual media-processing effort by 50% and automated video composition transitions, reducing visible flickering by 40%.
- Developed deployable AI backend services containerized with Docker and deployed on Amazon EC2 and Amazon S3.
- Contributed to quantitative forecasting workflows using XGBoost and Random Forest, achieving 15% lower forecasting error.

WorldQuant BRAIN

May 2025 – Jun 2026

Research Consultant (Part-time Remote)

- Designed and backtested 20+ quantitative alpha strategies across global equity datasets with multi-regime risk analysis.
- Applied time-series analysis, statistical validation, and drawdown limits, achieving Sharpe ratios consistently above 2.0.

FEATURED PROJECTS

Multi-Agent Claim Processing AI System [LangGraph, FastAPI, OCR, LLMs, Docker, Python]

- Engineered an agentic document intelligence workflow transforming unstructured insurance claim PDFs into structured JSON with schema validation, intelligent routing, and automated claim processing.

Financial AI Assistant (RAG) [RAG, LangChain, FAISS, FastAPI, Semantic Search]

- Developed a sub-second retrieval-augmented generation platform for complex financial reports using dense vector indexing, contextual chunking, and source attribution.

ADGM-Compliant Corporate Agent with Document Intelligence [LangChain, FastAPI, Python, Document Intelligence, OCR]

- Built a regulatory compliance agent parsing statutory filings and evaluating legal submissions against ADGM guidelines with traceable audit logs.

Automatic Number Plate Recognition System [YOLOv8, OpenCV, OCR, PyTorch, Python]

- Trained and benchmarked a near-real-time license plate detection and character extraction pipeline evaluated on 1,000+ annotated traffic frames across adverse conditions.

Logistics AI Tracker [React, FastAPI, LangChain, RAG, Docker, GitHub Actions]

- Created a full-stack supply chain tracking platform featuring a natural-language copilot for shipment incident queries with containerized CI/CD deployment.

TECHNICAL SKILLS

Programming: Python, SQL, JavaScript, C++

Generative AI & Agents: LLMs, RAG, LangChain, LangGraph, Agentic Workflows, Multi-Agent Orchestration, Prompt Engineering, Tool Calling

Models & Platforms: AWS Bedrock, Amazon Nova, Gemini, Hugging Face, Whisper Large v3, OpenAI APIs, Anthropic APIs, Ray 2

Retrieval & Vector Search: FAISS, Chroma, Vector Databases, Dense Embeddings, Semantic Search, HF Embeddings

Backend & Cloud: FastAPI, Node.js, REST APIs, Async Processing, Docker, Amazon EC2, Amazon S3, GitHub Actions, CI/CD, Linux

Machine Learning: Scikit-learn, PyTorch, XGBoost, Random Forest, Time-Series Forecasting, Statistical Validation, Model Evaluation

Multimodal AI: Speech-to-Text, OCR, Video Processing, FFmpeg, Text-to-Video, Video Generation, Structured Data Parsing

HONORS & LEADERSHIP

- International Quant Championship 2025: Global Rank 156 in Stage 1, conducted by WorldQuant.
- Flipkart GRID 6.0: National Semi-Finalist (Top 1% selected among 10,000+ engineering teams in India).
- All India Inter-NIT Yoga Championship: Second Runner Up (National Podium across all NITs).
- Additional Secretary, Mathematical Society, NIT Warangal (2022 – 2025).
- Event Coordinator, SpringSpree Cultural Festival, NIT Warangal (2021 – 2025).